ETALE SHEAVES DESCEND ACROSS A PROPER MODIFICATION THAT IS AN ISOMORPHISM OFF Z

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 887-905 of sources/stacks-spaces-pushouts/pinned-original-source.tex; proof span is lines 907-950, the complete author proof (42 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 12 of queue_spaces-pushouts.json (pb=1892, stmt 887-905, proof 907-950).

Editorial note. spaces-pushouts wave-1: fresh adjudication accept (2026-10-14); six-check dispatched pool (§2 row appended 02:4x).

Editorial note. Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'lemma-glue-etale-sheaf-modification' also exists in stacks-etale-cohomology — a different proposition in a different chapter file.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 003R: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 950–950. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GFW: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 817–828. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $f : Y' \to Y$ be a proper morphism of algebraic spaces over S. Let $i : Z \to Y$ be a closed immersion. Set $E = Z \times_Y Y'$. Picture*

$$\begin{array}{ccc} E & \xrightarrow[j]{} & Y' \\ {\scriptstyle g}\downarrow & & \downarrow{\scriptstyle f} \\ Z & \xrightarrow{i} & Y \end{array}$$

Date: stacks-project:spaces-pushouts.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

If f is an isomorphism over $Y \setminus Z$, then the functor

$$Sh(Y_{étale}) \longrightarrow Sh(Y'_{étale}) \times_{Sh(E_{étale})} Sh(Z_{étale})$$

is an equivalence of categories.

Proof. Observe that $X = Y' \coprod Z \to Y$ is a proper surjective morphism. Thus it suffice to construct an equivalence of categories

$$Sh(Y'_{étale}) \times_{Sh(E_{étale})} Sh(Z_{étale}) \longrightarrow \text{descent data for étale sheaves wrt } \{X \to Y\}$$

compatible with pullback functors from Y because then we can use Lemma 0GFW to conclude. Thus let $(\mathcal{G}', \mathcal{G}, \alpha)$ be an object of $Sh(Y'_{étale}) \times_{Sh(E_{étale})} Sh(Z_{étale})$ with notation as in Categories, Example 003R. Then we can consider the sheaf $\mathcal{F}$ on X defined by taking $\mathcal{G}'$ on the summand Y' and $\mathcal{G}$ on the summand Z. We have

$$X \times_Y X = Y' \times_Y Y' \amalg Y' \times_Y Z \amalg Z \times_Y Y' \amalg Z \times_Y Z = Y' \times_Y Y' \amalg E \amalg E \amalg Z$$

The isomorphisms of the two pullbacks of $\mathcal{F}$ to this algebraic space are obvious over the summands E, E, Z. The interesting part of the proof is to find an isomorphism $\mathrm{pr}_{0,small}^{-1}\mathcal{G}' \to \mathrm{pr}_{1,small}^{-1}\mathcal{G}'$ over $Y' \times_Y Y'$ satisfying the cocycle condition. However, our assumption that $Y' \to Y$ is an isomorphism over $Y \setminus Z$ implies that

$$h : Y \coprod E \times_Z E \longrightarrow Y' \times_Y Y'$$

is a surjective proper morphism. (It is in fact a finite morphism as it is the disjoint union of two closed immersions.) Hence it suffices to construct an isomorphism of the pullbacks of $\mathrm{pr}_{0,small}^{-1}\mathcal{G}'$and $\mathrm{pr}_{1,small}^{-1}\mathcal{G}'$ by h_{small} satisfying a certain cocycle condition. For the diagonal, it is clear how to do this. And for the pullback to $E \times_Z E$ we use that both sheaves pull back to the pullback of $\mathcal{G}$ by the morphism $E \times_Z E \to Z$. We omit the details. □